

AI-Driven Skill Gap Analysis and Career Development Framework Using Knowledge Graphs and Adaptive Interview Intelligence

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Abstract—The persistent misalignment between academic preparation and evolving industry skill demands presents a critical challenge in employability assessment and career guidance. Undergraduate graduates frequently enter the workforce without the practical competencies and specialized technical skills required by modern industries. This research proposes an integrated employability intelligence framework that combines graph-based skill gap analysis, personalized learning recommendations, skill validation, and adaptive interview assessment to provide a comprehensive evaluation of candidate readiness for industry roles. The system models the employability ecosystem using a heterogeneous knowledge graph that represents relationships among candidates, skills, job roles, learning resources, and projects. A Graph Neural Network (GNN) is employed to predict missing skills through link prediction and compute a continuous readiness score for target roles. Based on the identified skill gaps, the framework generates personalized learning paths and skill-conditioned capstone project recommendations using a teacher–student knowledge distillation approach with LoRA-based fine-tuning. In addition to skill development, the framework incorporates a transcript-based skill validation module that evaluates academic evidence through evidence-weighted scoring and AI-generated quizzes. To further assess practical readiness, an adaptive interview intelligence module simulates job interviews using retrieval-augmented generation and evaluates responses using semantic relevance and speech emotion recognition (SER) models. The integration of these components forms a unified system capable of dynamic employability assessment, targeted skill development guidance, and preparation for real-world recruitment processes.

Keywords—*Knowledge Graph, Graph Neural Networks, Skill Gap Analysis, Personalized Learning Path, Skill Validation, Interview Intelligence, Employability Assessment, Speech Emotion Recognition*

I. INTRODUCTION

The need for specialized technical skills has expanded dramatically due to the digital transformation of many industries. However, many graduates entering the workforce often lack the competencies required by modern employers, leading to

a persistent gap between academic preparation and industry expectations. Traditional academic evaluation methods, such as grades and course completion, primarily measure theoretical knowledge and do not fully capture a candidate’s practical abilities or readiness for real-world job roles. As a result, organizations have difficulty accurately evaluating candidates’ competencies, and students find it challenging to identify the abilities required to remain competitive in the labor market. Recent studies emphasize the need to close the skills gap between industry and academia with intelligent systems that can analyze both academic evidence and industry skill requirements in a data-driven manner [1].

Current employability analysis techniques usually depend on simple similarity-based skill comparisons or keyword matching between candidate profiles and job descriptions. These methods frequently fall short of capturing the intricate connections between roles, abilities, and educational materials. Knowledge Graphs (KGs) provide a structured approach for modeling these relationships by representing entities such as candidates, skills, and job roles as interconnected nodes. This representation enables multi-hop reasoning to identify hidden competency gaps [2].

GNNs further enhance this capability by learning representations from graph-structured data and capturing complex relational dependencies through message passing mechanisms, making them effective for tasks such as link prediction, skill-gap identification, and readiness assessment [3]. In addition, personalized recommendation techniques have been widely used to guide students toward relevant learning resources that address identified skill gaps and support career development [4].

Beyond identifying skill gaps, evaluating employability also requires verifying skill proficiency and assessing real-world job readiness. Academic transcripts provide valuable evidence of student learning, but translating course performance into

industry-relevant skills requires structured skill taxonomies and evidence-based scoring mechanisms. Furthermore, advances in Retrieval Augmented Generation (RAG) allow large language models to generate context-aware content by retrieving relevant knowledge from external sources. This capability enables the generation of adaptive interview questions tailored to specific job roles [5]. Such technologies can support realistic interview simulations that evaluate both technical knowledge and communication ability.

In addition to textual evaluation, communication skills and emotional expression play an important role in real recruitment processes. Recent advances in Automatic Speech Recognition (ASR) and SER enable automated analysis of spoken responses. These technologies allow systems to evaluate not only the semantic relevance of interview answers but also emotional confidence and delivery quality. Deep learning models such as Wav2Vec2 have demonstrated strong performance in extracting acoustic features and identifying emotional states from speech signals [6]. Integrating these capabilities into an employability assessment framework enables a more comprehensive evaluation of candidate readiness.

Inspired by these developments, this study proposes an integrated employability analysis framework that combines graph-based skill gap analysis, personalized industry-based project and learning recommendations, skills validation, and adaptive interview assessment. The framework models the employability ecosystem using a heterogeneous knowledge graph that connects candidates, skills, job roles, learning resources, and projects. A GNN is used to predict missing skills and calculate role-specific readiness scores. Based on the identified skill gaps, the framework generates personalized learning paths and skill-conditioned capstone project recommendations using a teacher–student knowledge distillation approach with LoRA-based fine-tuning, enabling efficient domain-specific project generation.

In addition, a transcript-based skill validation module verifies competencies using evidence-based scoring and AI-generated quizzes, while an adaptive interview intelligence framework simulates job interviews and evaluates responses using semantic matching and SER models. By integrating these components, the proposed framework provides a comprehensive and interpretable assessment of employability readiness.

The main contributions of this work are summarized as follows:

- A heterogeneous knowledge graph framework for modeling relationships among candidates, skills, job roles, learning resources, and projects in the employability domain.
- A Graph Neural Network based skill gap prediction approach for computing role-specific readiness scores and identifying missing competencies.
- A personalized learning path and capstone project recommendation mechanism that supports targeted skill development.

- An evidence-based skill validation module that combines transcript-derived academic evidence with AI-generated knowledge assessments.
- An adaptive interview intelligence system that evaluates both semantic response quality and emotional delivery using speech-based analysis.
- A unified employability intelligence pipeline that integrates skill analysis, validation, learning guidance, and interview readiness assessment into a single interpretable framework.

II. LITERATURE REVIEW

The growing gap between academic education and industry skill requirements has attracted significant research attention. Studies indicate that graduates often possess theoretical knowledge but lack the practical competencies demanded by modern employers. For example, Jaiswal et al. analyzed the relationship between artificial intelligence curricula and job market requirements and identified several mismatches between university training and industry expectations. Their findings highlight the need for systems capable of systematically identifying skill gaps and guiding learners toward relevant competencies [7]. In addition to analytical approaches, application-driven solutions have also been proposed to address this issue. For instance, an interactive application for university students was developed to reduce the industry–academia skill gap in the software engineering domain by providing skill tracking and career guidance features [8].

KGs have emerged as an effective approach for modeling complex relationships among entities such as skills, job roles, courses, and candidates. Hogan et al. describe KGs as structured representations of real-world knowledge that enable relational reasoning across interconnected entities [2]. In practical implementations, graph databases such as Neo4j have been widely adopted to efficiently manage and query large-scale KGs. For instance, Perera et al. demonstrated the effectiveness of Neo4j in building a legal information retrieval system, where graph-based representations significantly improved the retrieval and analysis of interconnected data [9].

Building upon these representations, GNN enable machine learning models to learn representations directly from graph structures. Through relational message passing, GNNs capture contextual dependencies and predict missing relationships within large-scale networks [3]. These capabilities make graph-based approaches particularly suitable for employability analysis and skill gap prediction.

Graph-based approaches have been used in several studies for hiring and career advice tasks. Although it does not describe multi-relational connections or provide skill-gap prediction, SkillRec [10] employs embedding-based approaches for skill recommendation. HRGraph [11] leverages KGs and large language models (LLMs) for job recommendation, focusing on data extraction and matching rather than predictive analysis or readiness estimation. A knowledge graph-based learning route system is proposed by Liu et al. [12], however it lacks candidate-specific skill modeling and functions at the

concept level. In contrast, a more thorough employability study is made possible by the suggested framework's usage of a heterogeneous GNN for link prediction in conjunction with role-aligned ranking, tailored learning, and validation modules.

Recent advances in LLMs have introduced new capabilities for knowledge-intensive reasoning and content generation. RAG combines parametric language models with external knowledge retrieval to produce context-aware responses and reduce hallucinations in generated outputs [5]. This technique has been widely adopted in applications such as question generation, conversational systems, and knowledge-grounded recommendations. Additionally, parameter-efficient fine-tuning methods such as Low-Rank Adaptation (LoRA) allow LLMs to be adapted for domain-specific tasks while updating only a small fraction of model parameters, significantly reducing computational cost [13].

Another important aspect of employability assessment is evaluating communication and interview performance. Shen et al. proposed a joint learning framework for intelligent job interview assessment that analyzes candidate responses relative to job requirements and supports interview question recommendation [14]. Furthermore, advances in SER have enabled systems to analyze emotional states and communication confidence from speech signals. Deep learning models such as Wav2Vec-based architectures have demonstrated strong performance in extracting acoustic features and identifying emotional states during spoken interactions.

Although these studies provide valuable contributions across skill analysis, recommendation systems, language models, and interview assessment, most existing approaches address these tasks independently. Few systems integrate graph-based skill gap prediction, personalized learning recommendations, project-based competency development, transcript-based skill validation, and adaptive interview evaluation within a unified framework. To address this limitation, the proposed research introduces an integrated employability intelligence framework that combines knowledge graph modeling, GNN-based skill prediction, learning path recommendation, AI-generated skill validation, project recommendation, and adaptive interview intelligence to provide a comprehensive and explainable assessment of candidate readiness.

III. METHODOLOGY

The proposed framework integrates graph-based skill prediction with transcript-driven validation and interview assessment to provide a comprehensive employability analysis. It consists of two main components: a KG-GNN module for skill-gap identification and recommendation, and a transcript-grounded module for skill validation and interview evaluation. Together, these components enable an end-to-end assessment of candidate readiness.

A. Knowledge Graph and GNN Framework for Personalized Learning and Project Recommendation

This study proposes a heterogeneous graph-based employability intelligence framework that integrates a KG, GNN, and

recommendation mechanisms to identify skill gaps and generate personalized learning paths and project recommendations.

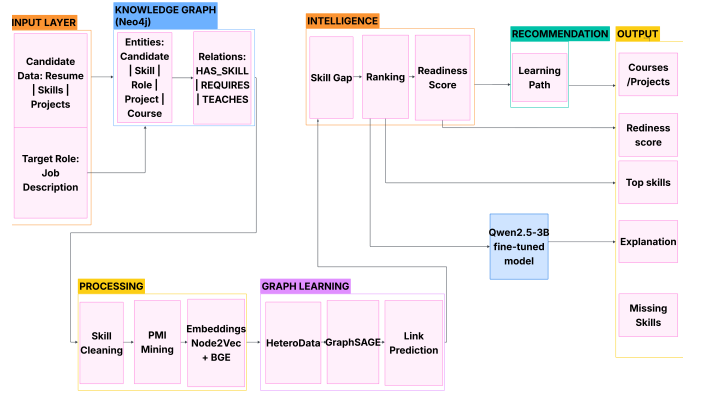


Fig. 1. Overview of the proposed heterogeneous graph-based employability intelligence framework.

1) *Knowledge Graph Construction*: A large-scale heterogeneous KG was constructed to represent the employability ecosystem and was implemented using the Neo4j graph database, which supports a property graph model for efficient storage, querying, and traversal of highly connected data. The final graph consists of 20,256 nodes and 114,687 relationships, with an average node degree of 11.32. The graph integrates 2,100 candidate profiles, 1,624 raw skills reduced to 396 canonicalized skills, 1,175 job postings, 925 learning resources, and 60 skill clusters.

Additionally, 7,995 *SIMILAR_TO* skill-skill edges were created to capture semantic and structural proximity between skills. The schema includes 17 node types and 21 relationship types such as *HAS_SKILL*, *REQUIRES_SKILL*, *WORKED_ON*, *USES_TECHNOLOGY*, *TARGETS_ROLE*, *TEACHES_SKILL*, *CO_OCCURS_WITH*, and *SIMILAR_TO*. Each relationship contains statistical attributes including support, confidence, and importance weights. This multi-relational schema enables multi-hop reasoning across candidates, roles, projects, and learning resources.

A partial visualization of the constructed knowledge graph in Neo4j is shown in Fig. 2.

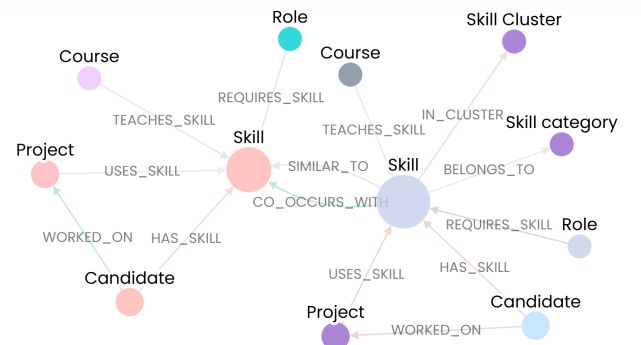


Fig. 2. Partial visualization of the heterogeneous knowledge graph implemented in Neo4j.

2) *Skill Canonicalization and Skill Relationship Mining*: Skill canonicalization was performed to reduce redundancy by mapping variant skill names into standardized canonical representations. For example, “Python programming” was normalized to “Python”.

Skill–skill relationships were computed using multi-source Pointwise Mutual Information (PMI) derived from job postings, projects, courses, and work experience data. The PMI between skills s_i and s_j is calculated as shown in (1):

$$PMI(s_i, s_j) = \log \frac{P(s_i, s_j)}{P(s_i)P(s_j)} \quad (1)$$

Edges were filtered using source-aware thresholds to remove weak associations. This process produced a structural skill topology with high-confidence connections suitable for relational learning.

3) *Hybrid Skill Embedding*: Using a hybrid embedding approach, skill representations were produced. Node2Vec was used to capture graph topology in structural embeddings, and the transformer-based BGE-large model was used to create semantic embeddings.

By combining these embeddings, rich feature vectors that represent both semantic similarity and structural relationships were created.

4) *Graph Export to PyTorch Geometric*: The Neo4j knowledge graph was converted into a PyG Geometric HeteroData structure to train graph neural networks. Every kind of relationship was transformed into an edge representation, and every kind of node was assigned a unique index.

Node features were constructed as follows:

- Skill nodes: embedding vectors representing semantic meaning
- Person nodes: numerical features such as number of skills, projects, and experience duration
- Project nodes: aggregated skill representations
- SkillCategory nodes: aggregated category-level features

This heterogeneous representation allows the model to capture both semantic and structural properties of the graph.

5) *Problem Formulation: Link Prediction*: The missing-skill detection task was formulated as a link prediction problem, where the goal is to predict whether a relationship exists between a Person node and a Skill node. The positive samples were obtained from already available HAS_SKILL edges, while the negative samples were derived by matching individuals with skills they lacked. This makes supervised learning possible for forecasting candidate-skill relationships.

6) *GNN Model Architecture and Training*: Using multi-relational message forwarding, a heterogeneous GNN (based on GraphSAGE) is used to learn node representations. In this system, each Person node collects contextual information from a variety of linked entities, including skill categories, related skills via *SIMILAR_TO* and *CO_OCCURS_WITH* relationships, possessed talents, and related projects. Similarly, by capturing structural dependencies within the skill graph, skill nodes allow the model to learn meaningful representations of relationships throughout the employability ecosystem.

The model is trained using the Adam optimizer with a learning rate of 0.001 and dropout regularization to reduce overfitting. It has 2-3 layers with hidden dimensions between 128 and 256. The model is trained using binary cross-entropy loss to differentiate between relationships that exist and those that do not. Negative sampling is applied using a 1:2 ratio, where two negative samples are created for each positive edge. The following is the definition of the expected probability of a person-skill relationship:

$$P_{\text{gnn}}(p, s) = \sigma(f(h_p, h_s)) \quad (2)$$

where h_p and h_s are node embeddings.

7) *Role-Aligned Skill Ranking*: Predicted skills were filtered based on the required skill set of the target role. Skills already possessed by the candidate were removed. Remaining skills were ranked using a role-conditioned scoring function that combines GNN prediction probability with role importance derived from TF-IDF analysis of job descriptions.

The final ranking score is defined as:

$$Score(s) = P_{\text{gnn}}(p, s) \times Importance(r, s) \quad (3)$$

8) *Personalized Learning Path Recommendation*: The system generates a personalized learning path by mapping prioritized missing skills to courses through the *TEACHES_SKILL* relationships within the knowledge graph. Courses are ranked based on their coverage of high-priority missing skills and overall gain score.

9) *Capstone Project Recommendation*: To translate skill gap analysis into demonstrable competencies, the framework includes a capstone project recommendation module. The GNN-generated missing skill rankings and readiness scores are serialized into structured inputs and provided to a language model.

Project recommendations are generated using high-capacity teacher models from the Gemini and Qwen families. A smaller student model (Gemma 3 4B) is trained using a teacher–student knowledge distillation paradigm to produce domain-specific project recommendations.

Low-Rank Adaptation (LoRA) enables efficient fine-tuning by updating only 2.7% of model parameters.

10) *Performance Evaluation and Quantitative Validation*: The closed-loop mechanism is validated as an iterative, measurable process: expert feedback is collected, rating distributions are summarized, recurring themes are extracted, prompt revisions are generated under explicit constraints, and updated prompts are version-tracked for subsequent regeneration cycles. This enables repeatable improvement experiments with auditable artifacts across prompt versions.

Evaluation is conducted with a structured multi-dimensional rubric (Likert 1-5) over five quality dimensions: skill-gap accuracy, project relevance, tech-stack appropriateness, implementation-step quality, and overall quality.

Algorithm 1 identifies recurring weaknesses and strengths by combining per-dimension rating statistics with LLM-based

analysis of expert comments, then compiles them into a structured pattern report.

Algorithm 1 Pattern Detection Algorithm

Require: Feedback entries $F = \{f_1, f_2, \dots, f_N\}$
Ensure: Pattern report R

```

1:  $S \leftarrow \text{ComputeRatingStatistics}(F)$  ▷ Per-dimension stats
2:  $W \leftarrow S.\text{weak\_dimensions}$ 
3:  $P \leftarrow S.\text{strong\_dimensions}$ 
4: if HasFreeTextComments( $F$ ) then
5:    $C \leftarrow \text{Concatenate}(\{f_i.\text{free\_text\_comments} \mid f_i \in F\})$ 
6:    $\text{prompt} \leftarrow \text{FormatAnalysisPrompt}(S, C)$ 
7:    $T \leftarrow \text{ParseJSONOrFallback}(\text{LLM}(\text{prompt}))$ 
8: else
9:    $T \leftarrow \{\text{criticisms: ["No comments"], praises: [],}$ 
      $\text{recommendations: []}\}$ 
10: end if
11:  $A \leftarrow \text{ExtractMeanRatings}(S)$ 
12:  $R \leftarrow \text{BuildPatternReport}(|F|, A, W, P, T.\text{criticisms},$ 
      $T.\text{recommendations}, \text{JSONString}(T))$ 
13:  $\text{SavePatternReport}(R)$ 
14: return  $R$ 

```

11) Feedback-Driven Prompt Evolution: Rather than relying on token-level preference optimization pipelines, our system performs instruction-level refinement inside the distillation workflow. The refinement data is derived from comprehensive recommendation reviews, where each reviewed record consists of a single, detailed feedback entry provided by a domain expert. This core mechanism uses structured expert ratings and qualitative commentary to update the system prompt through a controlled meta-prompting step, while preserving required output structure and high-performing behaviors. This design keeps the refinement loop practical for iterative deployment and directly aligned with domain-expert judgment.

To ensure stable and reliable refinement outcomes, we apply a data-sufficiency policy before major prompt updates. In operational runs, we target at least 200 reviewed records for full refinement cycles, with smaller batches used for early diagnostics. This aligns with default large-batch setting for regeneration and iteration workflows, and supports statistically more credible pattern extraction at each cycle.

Algorithm 2 uses this pattern report to revise the current system prompt through meta-prompting and records the evolved prompt as a versioned update for the next training cycle.

Algorithm 2 Prompt Evolution via Meta-Prompting

Require: Current prompt p , pattern report R
Ensure: Evolution record E

```

1:  $\text{meta\_prompt} \leftarrow \text{BuildEvolutionPrompt}(p, R)$ 
2:  $p' \leftarrow \text{LLM}(\text{meta\_prompt})$ 
3:  $p' \leftarrow \text{StripMarkdownFences}(p')$ 
4:  $\Delta \leftarrow \text{GenerateDiffSummary}(p, p')$ 
5:  $v \leftarrow \text{GetCurrentPromptVersion}()$ 
6:  $v' \leftarrow \text{GetNextVersionString}()$ 
7:  $E \leftarrow \text{CreatePromptEvolution}(v, v', R.\text{id}, p, p', \Delta)$ 
8:  $\text{SavePromptEvolution}(E)$ 
9: return  $E$ 

```

12) Explainable AI Layer: To ensure interpretability, an explainable AI module translates structured model outputs into human-readable explanations. A structured explanation

dataset was constructed from graph-derived employability signals including readiness scores, matched skills, ranked missing skills, and project alignment metrics. Transformer models such as Qwen2.5-3B and DistilGPT-2 were fine-tuned to generate coherent explanations.

B. Transcript-Grounded Skill Validation and Adaptive Interview Intelligence Framework

The second component of the system introduces a transcript-based employability intelligence framework that integrates academic evidence extraction, skill estimation, adaptive validation, job matching, and quantitative interview assessment.

1) Transcript Parsing and Academic Evidence Extraction: Academic transcripts are accepted in PDF and PNG formats. Text is extracted using PDFPlumber for digital transcripts and Optical Character Recognition (OCR) for image-based documents. A rule-based extraction pipeline identifies course titles, grades, credit values, and academic year.

2) Course-to-Skill Mapping: The transcript-based skill validation module relies on a structured taxonomy of 65 skills developed through a combination of industry consultation, academic input, and analysis of emerging labor market trends in Sri Lanka. The primary objective of this taxonomy is to bridge the gap between academic curricula and industry skill requirements by aligning course-level learning outcomes with practical competencies expected in real-world job roles.

To ensure consistency and reduce redundancy, skill names were normalized into canonical representations. For example, variations such as “Python programming” and “Python development” were standardized to “Python.” The curated skill set was further validated against widely recognized frameworks, including reports from the World Economic Forum, and refined with input from subject matter experts to ensure relevance to contemporary industry demands.

Each skill was mapped to academic courses using curriculum data from multiple universities in Sri Lanka. This mapping captures the relationship between theoretical instruction and applied competencies, enabling the system to infer skill proficiency from transcript data. Courses that could not be confidently mapped to any skill were excluded or assigned zero contribution, ensuring that only validated evidence contributes to skill estimation.

Although the taxonomy is primarily designed for the Sri Lankan context, it has been structured to support adaptability across different educational systems. The core set of skills reflects broadly recognized competencies in domains such as software engineering, data science, and communication. Future extensions will incorporate region-specific curriculum data and industry requirements to improve generalizability across institutions and countries.

The taxonomy is organized into 15 skill categories, including programming languages (e.g., Python, Java, JavaScript), databases (e.g., SQL, PostgreSQL, MongoDB), web development (e.g., HTML, CSS, React), artificial intelligence and machine learning (e.g., TensorFlow, PyTorch, NLP), cloud

computing (e.g., AWS, Azure, GCP), DevOps and infrastructure (e.g., Docker, Kubernetes, CI/CD), data engineering (e.g., Apache Spark, ETL), mobile development (e.g., Android, iOS), software testing, data science and analytics (e.g., Power BI, Pandas), computer networks, cybersecurity, software development methodologies (e.g., Agile, Scrum), operating systems, and soft skills related to communication and employability.

3) *Evidence-Based Skill Scoring*: Skill mastery is estimated using an evidence-weighted aggregation model that integrates grade performance, course credits, skill relevance, and recency.

The recency factor adjusts the weight of older courses, ensuring that more recent courses contribute more to the skill score. The recency factor is defined as:

$$Recency_i = e^{-0.4 \times years_since_i} \quad (4)$$

In this formula, the exponent $e^{-0.4 \times years_since_i}$ was empirically chosen to give a reasonable decay over time. The $years_since_i$ refers to the number of years since the course was completed, and the decay factor ensures that recent courses have a stronger influence on the final skill score.

Next, the evidence weight for each course is calculated as follows:

$$EvidenceWeight_i = map_weight_i \times credits_i \times Recency_i \quad (5)$$

Here:

map_weight_i represents the relevance of the course to the skill, based on a predefined mapping between courses and skills. $credits_i$ reflects the number of academic credits awarded for the course, representing its relative importance in the curriculum. $Recency_i$ accounts for the time since the course was completed, as explained above.

The contribution from each course to the final skill score is then calculated as:

$$Contribution_i = grade_norm_i \times EvidenceWeight_i \quad (6)$$

Here, $grade_norm_i$ is the normalized grade for the course, which adjusts the contribution based on the student's performance.

Finally, the overall skill score is calculated by aggregating the contributions from all courses and normalizing them by the total evidence weight:

$$SkillScore = \frac{\sum Contribution_i}{\sum EvidenceWeight_i} \times 100 \quad (7)$$

This formula ensures that the final skill score takes into account the student's performance, the relevance and weight of each course, and the recency of the courses completed.

4) *Confidence Estimation*: To measure the reliability of the inferred skill score, a confidence score is computed. This score represents how strongly the system believes in the accuracy of the inferred skill score based on the amount of available evidence. The confidence score is calculated as:

$$Confidence = 1 - e^{-0.25 \times total_evidence} \quad (8)$$

In this formula: $total_evidence$ refers to the total evidence accumulated across all courses, taking into account the credits, course relevance, and recency factors.

The 0.25 coefficient was empirically chosen to provide a reasonable scale for confidence estimation. It reflects how much evidence is needed to confidently assign a high skill score. In future iterations, this value can be adjusted based on further calibration or domain-specific data.

The coefficient 0.25 was selected to ensure that even with limited evidence (e.g., only a few courses mapped to a skill), the confidence score does not become too high prematurely, while still allowing confidence to increase significantly as more evidence is accumulated.

5) *Skill Validation Using AI-Generated Quizzes*: The system validates inferred skills through questionnaire-based assessments. Questions are generated using a RAG pipeline integrated with a ChromaDB vector store. Generated questions are stored in a question bank and dynamically selected during evaluation.

6) *Job Matching*: Candidate skill vectors are compared with job requirement vectors using cosine similarity. TF-IDF derived importance weights ensure that discriminative skills have greater influence on ranking.

7) *Interview Simulation*: When a candidate selects a job role, the job description is segmented into contextual blocks and embedded using a semantic embedding model. These embeddings are stored in a FAISS vector database.

Interview questions are generated using a RAG pipeline, which retrieves relevant contextual information and injects it into structured prompts, ensuring domain-specific and context-aware question generation.

8) *Speech-Based Response Analysis*: Candidate answers are recorded using voice input. An ASR module transcribes speech to text. SER is performed using transformer-based Wav2Vec2 model [15], which extracts contextual acoustic representations. Features such as pitch, energy, speaking rate, and spectral embeddings are used to infer emotional states such as confidence, stress, and neutrality.

9) *Interview Evaluation*: Each response is evaluated using multiple signals including semantic relevance (SR), emotional confidence (EC), response coherence (RC), and linguistic clarity (LC). The final interview score is computed as a weighted aggregation:

$$InterviewScore = 0.4 \times SR + 0.3 \times EC + 0.15 \times LC + 0.15 \times RC \quad (9)$$

The weights were empirically determined based on pilot experiments to prioritize technical correctness (semantic relevance) while maintaining importance of delivery and communication.

Overall, the proposed methodology integrates knowledge graph modeling, GNN-based skill prediction, learning path recommendation, transcript-grounded skill validation, AI-generated assessments, job matching, contextual interview

simulation, and speech-based evaluation into a unified employability intelligence framework.

10) End-to-End Interview Evaluation Protocol: To evaluate the effectiveness of the interview intelligence module beyond isolated SER model testing, a controlled pilot user study was conducted with 18 participants. Each participant completed a simulated interview session consisting of 5–7 questions generated for a selected target role. During each session, the system recorded average semantic relevance, average emotional confidence, overall interview score, and post-session user satisfaction.

The purpose of this pilot study was to examine whether the integrated interview pipeline, including context-aware question generation, speech transcription, semantic scoring, and emotion-aware response analysis, produced coherent and practically useful interview assessments. Since this evaluation was conducted as an initial system-level validation, the results are interpreted as preliminary evidence of integrated module consistency rather than as a definitive large-scale benchmark.

IV. RESULTS AND EVALUATION

A. Dataset Description and Experimental Setup

For the graph-based component, the dataset is constructed from the heterogeneous knowledge graph, where relationships between `Person` and `Skill` nodes are treated as positive samples for the link prediction task. Each edge between a person and a skill represents the presence of an observed skill in the individual’s profile.

Negative samples are generated randomly by pairing a person node with skills that are not present in their profile, following a 1:2 positive-to-negative sampling strategy.

An edge-level split is applied to divide the dataset into training, validation, and test sets with proportions of 70%, 15%, and 15%, respectively. This approach prevents information leakage and ensures that the model is trained, tuned, and evaluated on mutually exclusive edge sets.

To evaluate the model’s performance, several link prediction metrics are used on the test set, including Hits@K (for $K = 5, 10, 20$), Mean Reciprocal Rank (MRR), and Area Under the Curve (AUC).

The proposed method is compared against two baseline approaches: (i) an embedding similarity-based ranking method, and (ii) a popularity-based ranking strategy.

For the interview assessment component, the SER model is evaluated using the RAVDESS dataset [16], which contains labeled emotional speech samples across multiple categories. The dataset was divided into training, validation, and test sets using a 70%, 15%, and 15% split. The model was trained for 25 epochs with batch size of 16, using Adam optimizer with a learning rate of $1e-4$. Raw audio waveforms were used as input, and contextual acoustic features were extracted using Wav2Vec2 architecture. This experimental setup ensures a fair comparison between models and allows robust evaluation of emotion classification performance.

B. GNN-Based Skill Prediction Performance

The results demonstrate that the proposed GNN model effectively captures both structural and semantic relationships within the knowledge graph, consistently outperforming the embedding similarity baseline across all ranking-based metrics. In particular, improvements in Hits@10 and Hits@20 indicate that relevant missing skills are more frequently ranked among the top predictions.

The GNN model outperforms the popularity baseline in ranking-based metrics like Hits@K and MRR, demonstrating its effectiveness for generating personalized and role-aware skill suggestions. Although the popularity baseline achieves a marginally higher AUC (0.975) compared to the GNN model (0.971), the difference is relatively small and within a narrow margin. This suggests that both models have comparable global classification ability. However, AUC evaluates overall discrimination between positive and negative edges, whereas the primary objective of this work is to rank the most relevant missing skills at the top of the recommendation list.

As a result, ranking-based metrics such as MRR and Hits@K are more suited for assessing the efficacy of the suggested method. The superior performance of the GNN model in these metrics indicates its advantage in prioritizing actionable and role-relevant skill gaps.

TABLE I summarizes the performance comparison across all methods.

TABLE I
GNN SKILL PREDICTION PERFORMANCE

Method	H@5	H@10	H@20	MRR	AUC
Embedding Sim.	0.120	0.253	0.461	0.091	0.842
Popularity Base.	0.198	0.364	0.742	0.143	0.975
Proposed GNN	0.214	0.412	0.791	0.158	0.971

C. Interview Emotion Recognition Evaluation

The interview intelligence module evaluates candidate communication and emotional confidence during simulated interviews. To assess the effectiveness of the SER component, a baseline convolutional neural network (CNN) model was compared with a transformer-based Wav2Vec2 architecture.

The evaluation results indicate that the Wav2Vec2 model significantly outperforms the CNN baseline in emotion classification accuracy. The improvement can be attributed to the transformer architecture’s ability to capture contextual speech representations and long-range acoustic patterns.

In TABLE II, the results show the comparison between the CNN Baseline and the Wav2Vec2 Model:

TABLE II
INTERVIEW EMOTION RECOGNITION RESULTS

Model	Accuracy
CNN Baseline	62%
Wav2Vec2 Model	71%

Further demonstration of the classification capability of the Wav2Vec2 model is shown in Fig. 3. The confusion matrix

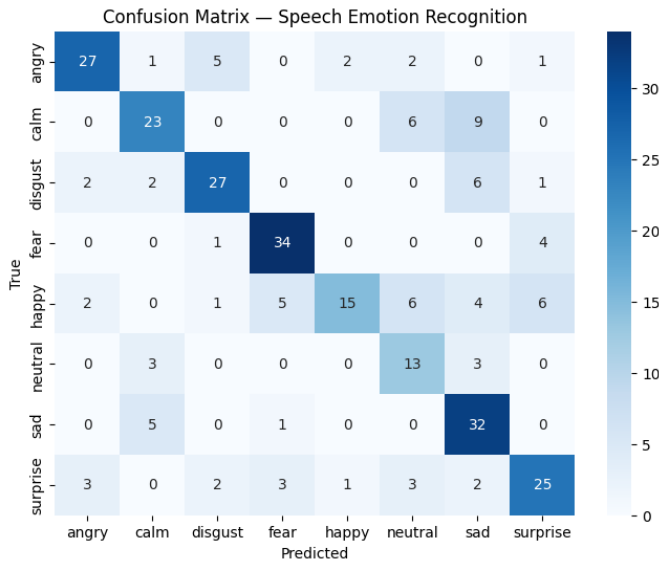


Fig. 3. Confusion matrix of speech emotion recognition using the Wav2Vec2 model.

indicates that the model is highly accurate in identifying emotions such as *fear*, *sad*, and *angry*. However, minor misclassifications occur when distinguishing between closely related emotions such as *happy* and *neutral*. Overall, the results indicate that the transformer-based approach effectively captures nuanced emotional patterns and is therefore suitable for evaluating communication behavior during interviews.

D. Project Recommendation and Explainability Results

The project recommendation module is analyzed in terms of parameter-efficient specialization and computational efficiency, rather than as a fully benchmarked optimization system. Using LoRA with rank 64, the framework fine-tuned only 2.70% of the parameters of the Gemma 3 4B student model while achieving domain-specific adaptation. Integration with the Unsloth framework further reduced memory and computational requirements compared with reinforcement learning-based approaches such as RLHF and DPO.

Embedding expert guidance directly into model weights also provides a practical alternative to static prompt-based systems by reducing reliance on long runtime instructions and supporting more efficient deployment.

To evaluate the explainability module, two language models were compared for natural-language explanation generation. DistilGPT2 achieved a BERTScore of 0.83, while Qwen2.5-3B-Instruct achieved a BERTScore of 0.89, indicating stronger coherence and alignment with structured employability signals. These results suggest that Qwen2.5-3B-Instruct produces more faithful explanations of graph-derived reasoning outputs.

E. Retrieval-Augmented Interview Generation

The RAG interview module was evaluated through a comparative analysis between baseline question generation without

retrieval and RAG-based question generation grounded in semantically retrieved job-description context. Context relevance was assessed using cosine similarity between embeddings of generated interview questions and the target job description, providing a quantitative measure of semantic alignment.

The results showed a substantial improvement in relevance, increasing from 0.33 without retrieval to 0.86 with RAG. This indicates that retrieval-grounded generation produces interview questions that are substantially better aligned with role-specific requirements. In addition to the quantitative improvement, qualitative inspection showed that the RAG-based approach reduced irrelevant questions and minimized unsupported technical content.

These findings suggest that the FAISS-based semantic retrieval mechanism effectively grounds the question generation process in job-specific context, resulting in more relevant and realistic interview simulations.

F. Preliminary End-to-End Pipeline Validation

Since the main contribution of this work lies in the integration of multiple employability analysis modules into a unified framework, a preliminary end-to-end validation was conducted to examine whether the outputs of each component remain consistent and actionable when executed as a complete pipeline. Rather than evaluating each module in isolation, this validation focused on the sequential flow from candidate profile ingestion to skill-gap prediction, role-aligned ranking, personalized learning path generation, project recommendation, transcript-grounded skill validation, and interview assessment.

Due to the current absence of large-scale longitudinal employability outcome data, the integrated framework was assessed through preliminary case-based pipeline analysis. The evaluation focused on four system-level criteria: (1) role relevance of predicted missing skills, (2) consistency between predicted skill gaps and recommended learning resources, (3) alignment between validated skills and interview focus areas, and (4) interpretability of the final recommendations and explanations.

The preliminary analysis showed that the framework maintained coherent transitions across modules. Skills predicted as missing by the KG-GNN module were reflected in the personalized learning path recommendations, while transcript-grounded validation and interview question generation remained consistent with the selected target role. In addition, the explainability layer produced human-readable justifications that linked graph-based predictions, validated competencies, and interview evaluation signals in an interpretable manner.

Although this preliminary validation does not yet constitute a full benchmark-based system evaluation, it provides evidence that the proposed architecture functions coherently as an integrated employability intelligence pipeline. A large-scale end-to-end evaluation with user studies, recruiter feedback, and longitudinal career outcome tracking is left for future work.

V. CONCLUSION AND FUTURE WORK

A. Conclusion

This research presents a unified employability intelligence framework that moves beyond traditional skill matching by integrating predictive, validation, and assessment capabilities into a single system. Rather than treating employability analysis as isolated tasks, the proposed approach combines knowledge graph reasoning, graph neural network-based skill prediction, and downstream recommendation and evaluation modules to enable structured and interpretable decision-making.

A key contribution of this work is the ability to bridge predicted competencies with validated evidence and real-world performance signals. The transcript-based validation and interview intelligence modules make sure that these predictions are supported by academic research and real-world evaluation, while the KG-GNN module finds role-specific skill gaps and produces actionable learning and project recommendations. Compared to stand-alone methods, this integration allows for a more accurate assessment of candidate preparedness.

Overall, the framework provides a scalable and interpretable foundation for intelligent career guidance systems that align academic preparation with industry expectations.

B. Future Work

Future work will focus on strengthening system-level integration by introducing unified evaluation metrics that quantify the alignment between predicted skill gaps, validated competencies, and interview performance. This will make it possible to evaluate end-to-end employability outcomes more thoroughly.

Furthermore, the framework could incorporate additional information from real-world recruitments, including hiring decisions and recruiter opinions, to evaluate and validate predictions produced by the system and further adjust the model accordingly. In addition, updates of the knowledge graph based on changes in industry trends and job market evolution could help to keep the model up to date and adaptable to changing conditions.

Another potentially useful extension is the integration of coding exercises and portfolio-based assessments alongside transcript and interview signals to provide a more comprehensive evaluation of practical skills. Additionally, the potential for collaboration with colleges and universities can be examined in order to obtain organized data from an academic

environment, including mapping of curricula and tracking of student performance on a continual basis, thus allowing for more precise skills verification using transcripts. Such collaboration will also assist in implementing the system in a real-world context, within educational institutions. Finally, deploying the framework in real-world educational or career platforms will enable longitudinal analysis of user progress and facilitate continuous refinement of recommendation and assessment strategies.

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